



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

ASSIGNMENT 2 DISCRETE STRUCTURE

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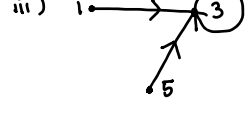
ASSIGNMENT 2 CHAPTER 2

Question 1 [$3x+y$ is multiple of 6]

- i) $x = \{1, 3, 5\}$ iv) Relation R is not asymmetric
 $y = \{1, 3, 5\}$ $(3, 3) \in R$

$$R = \{(1, 3), (3, 3), (5, 3)\}$$

- ii) Domain of $R = \{1, 3, 5\}$
 Range of $R = \{3\}$



Question 2

$$A = \{x, y, z\}, (x, y) \in R, (y, z) \in R$$

R is an equivalence relation

$\therefore$ R is reflexive, transitive and symmetric.

R is reflexive:

$$(x, x), (y, y), (z, z) \in R$$

R is transitive:

$$(x, y) \in R \wedge (y, z) \in R \rightarrow (x, z) \in R$$

R is symmetric:

$$\text{Since } (x, y) \in R \rightarrow (y, x) \in R$$

$$(y, z) \in R \rightarrow (z, y) \in R$$

$$(x, z) \in R \rightarrow (z, x) \in R$$

$$R = \{(x, x), (x, y), (x, z), (y, y), (y, x), (y, z), (z, z), (z, x), (z, y)\}$$

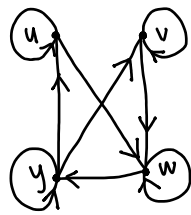
Question 3

$$i) B = \{u, v, w, y\}$$

$$R = \{(u, u), (u, w), (v, v), (v, w), (w, w), (w, y), (y, u), (y, v), (y, y)\}$$

$$M_R = \begin{matrix} & \begin{matrix} u & v & w & y \end{matrix} \\ \begin{matrix} u \\ v \\ w \\ y \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix} \end{matrix}$$

- ii) in-degrees: $u: (u, u), (y, u) \Rightarrow 2$
 $v: (v, v), (y, v) \Rightarrow 2$
 $w: (u, w), (v, w), (w, w) \Rightarrow 3$
 $y: (w, y), (y, y) \Rightarrow 2$



out-degrees: $u: (u, u), (u, w) \Rightarrow 2$

$v: (v, v), (v, w) \Rightarrow 2$

$w: (w, w), (w, y) \Rightarrow 2$

$y: (y, u), (y, v), (y, y) \Rightarrow 3$

- iii) Partial order: reflexive, antisymmetric, transitive

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ reflexive

$(u, w) \in R$ but $(w, u) \notin R$

$(v, w) \in R$ but $(w, v) \notin R$

$(w, y) \in R$ but $(y, w) \notin R$

$(y, u) \in R$ but $(u, y) \notin R$

$(y, v) \in R$ but $(v, y) \notin R$

$\rightarrow$ antisymmetric

$(u, w), (w, y) \in R$ but $(u, y) \notin R$

$\Rightarrow$ not transitive

$\therefore$ relation R is not a partial order relation

Question 4

$$f(x) = (x-1)^2, \quad x \in [1, \infty)$$

$$f(x_1) = f(x_2)$$

$$(x_1-1)^2 = (x_2-1)^2$$

$$\sqrt{(x_1-1)^2} = \sqrt{(x_2-1)^2}$$

$$|x_1-1| = |x_2-1|$$

Since $x_1, x_2 \in [1, \infty)$

$$x_1-1 \geq 0 \rightarrow |x_1-1| = x_1-1$$

$$x_2-1 \geq 0 \rightarrow |x_2-1| = x_2-1$$

$$|x_1-1| = |x_2-1|$$

$$x_1-1 = x_2-1$$

$$x_1 = x_2$$

$\therefore$ one-to-one function

$$\text{Let } f(x) = y$$

$$\forall y \in [0, \infty), \exists x \in [1, \infty)$$

$$y = (x-1)^2 \quad x \in [1, \infty)$$

$$\sqrt{y} = x-1$$

$$\text{since } x-1 \geq 0, \sqrt{(x-1)^2} = \sqrt{y}$$

$$\sqrt{y} = x-1$$

$$x = 1 + \sqrt{y}$$

$$y \geq 0, \sqrt{y} \geq 0$$

$\therefore x = 1 + \sqrt{y} \geq 1$ which is in the domain ($x \in [1, \infty)$)

$\therefore$ onto function

$\therefore$ Since f is both one-to-one and

onto function, f is bijective

Question 5

$$f(x) = 9x + 4, \quad g(x) = \frac{3}{2}x - 1$$

$$a) g^{-1}(y) = x$$

$$y = \frac{3}{2}x - 1$$

$$y+1 = \frac{3}{2}x$$

$$\frac{2}{3}(y+1) = x$$

$$g^{-1}(y) = \frac{2}{3}(y+1)$$

$$b) (g \circ f)(x) = g(f(x))$$

$$= \frac{3}{2}(9x+4) - 1$$

$$= \frac{27}{2}x + 6 - 1$$

$$= \frac{27}{2}x + 5$$

$$c) (f \circ g)(x) = f(g(x))$$

$$= 9\left(\frac{3}{2}x - 1\right) + 4$$

$$= \frac{27}{2}x - 9 + 4$$

$$= \frac{27}{2}x - 5$$

$$d) (f \circ g \circ g)(x) = f(g(g(x)))$$

$$= f\left(\frac{3}{2}\left(\frac{3}{2}x - 1\right) - 1\right)$$

$$= f\left(\frac{9}{4}x - \frac{3}{2} - 1\right)$$

$$= f\left(\frac{9}{4}x - \frac{5}{2}\right)$$

$$= 9\left(\frac{9}{4}x - \frac{5}{2}\right) + 4$$

$$= \frac{81}{4}x - \frac{45}{2} + 4$$

$$= \frac{81}{4}x - \frac{37}{2}$$

Question 6

$$a) P_0 = 4.0$$

$$P_1 = 5.0$$

$$P_t = P_{t-1} + \frac{1}{4}P_{t-2}, \quad t \geq 2$$

$$b) P_0 = 4.0$$

$$P_1 = 5.0$$

$$P_2 = 5.0 + \frac{1}{4}(4.0) = 6.0$$

$$P_3 = 6.0 + \frac{1}{4}(5.0) = 7.25$$

$$P_4 = 7.25 + \frac{1}{4}(6.0) = 8.75$$

$$P_5 = 8.75 + \frac{1}{4}(7.25) = 10.5625$$

Question 7

$$a) s(n) \{$$

if $(n=1)$

return 2

else

return $(s(n-1) * s(n-1)) - 1$

}

$$b) s(4)$$

$$n = 4$$

$$\text{return } (s(3) * s(3)) - 1$$

$\downarrow$

$$s(3)$$

$$n = 3$$

$$\text{return } (s(2) * s(2)) - 1$$

$\downarrow$

$$s(2)$$

$$n = 2$$

$$\text{return } (s(1) * s(1)) - 1$$

$\downarrow$

$$s(1)$$

$$n = 1$$

return 2

$$s(4) = 63$$

$$\text{return } (6 \times 6) - 1 = 63$$

$$s(3) = 8$$

$$\text{return } (3 \times 3) - 1 = 8$$

$$s(2) = 3$$

$$\text{return } (2 \times 2) - 1 = 3$$

$$s(1) = 2$$

return 2

$\therefore$ Answer = 63